TECHNICAL CASE STUDY / NOVEMBER 2026

Offline Field Library

Architecture, implementation choices, and operating model

A portable local knowledge system for emergency and technical references

Prepared for peer review by engineers interested in agentic coding.

THE PROJECT IN ONE MINUTE

A local web dashboard plus a large offline corpus, designed to travel on an external SSD.

The browser interface, search index, Kiwix server, and reference files are kept as distinct parts. The application runs on localhost; reference documents stay in category folders, and Wikipedia is read from a Kiwix ZIM archive.

1 TB	**119 GiB**	**81**	**14**
marketed SSD capacity	English Wikipedia maxi ZIM	reference files, excluding source notes	dashboard categories

WHY THIS REPORT EXISTS

It records the decisions and build sequence so a teammate or future AI coding agent can understand not only what is on the drive, but why the boundaries, dependencies, and update procedures were chosen. The companion PROJECT HANDOFF.md is the operational reference stored at the library root.

Scope note: point-in-time snapshot. Filenames, package versions, and free space can change; recheck the drive before maintenance.

01 / PROBLEM FRAMING

Requirements shaped the architecture

The user wanted a portable emergency and general-reference library, usable through Safari or Chrome without an internet connection.

Requirement	Implementation response	Reason
Offline operation	Static browser UI, local Python server, local files, local SQLite index, local Kiwix server.	No cloud service or runtime web API is needed after content is downloaded.
Low dependency count	Vanilla HTML/CSS/JS; Python standard library; SQLite FTS5; vendored pypdf.	Fewer services to install, update, or troubleshoot in an emergency.
Content can grow	Application in app/; references in content/<category>/; index in index/.	Content can be copied and audited separately from application code.
Large encyclopedia	Kiwix serves the English Wikipedia ZIM on port 8081.	ZIM is a purpose-built offline web archive; it is not copied into the dashboard's text index.
Search PDFs	pypdf extracts embedded text; SQLite FTS5 stores text and makes snippets.	Useful with no network or remote search engine.
Cross-device start	Platform-specific launchers and bundled runtime archives for Mac Intel/ARM and Windows x64.	No system-wide Python install is needed on target devices.

AS-BUILT SNAPSHOT

Data volume	State at documentation time
Reference corpus	About 120 GB on disk, dominated by the 119 GiB English Wikipedia ZIM; 81 non-source-note reference files across categories.
Application and packages	About 505 MB in app/; compressed runtime packages are around 136 MB. SQLite index is about 7 MB and rebuildable.
Drive headroom	The 1 TB SanDisk SSD is exFAT. macOS last reported about 762 GiB free; verify again before future large downloads.
Web endpoints	Dashboard at 127.0.0.1:8765; Kiwix at 127.0.0.1:8081.

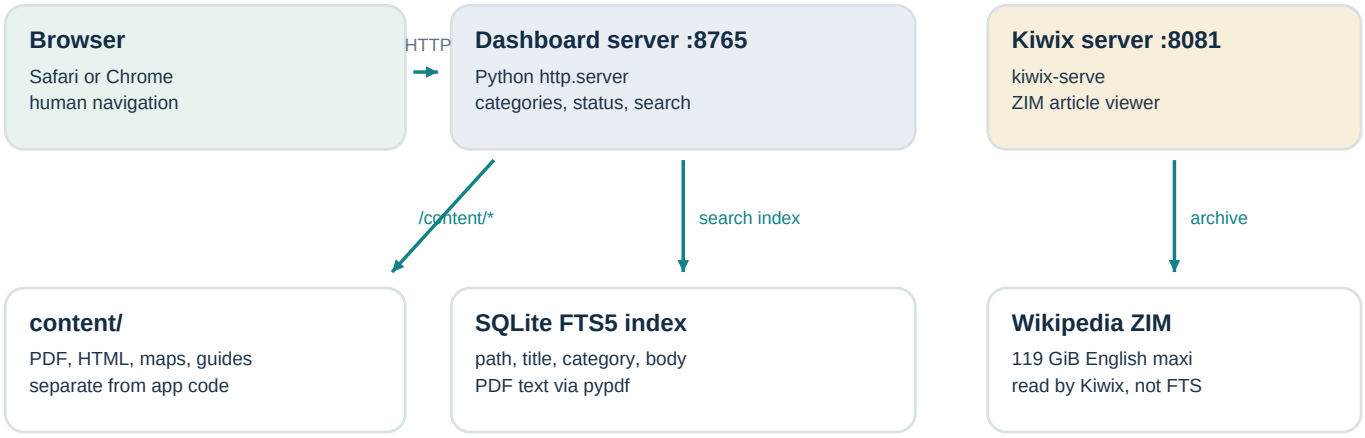
Offline has a boundary

Saved files and captured page text work offline. Links to live alerts, current shelter lists, online maps, manufacturer portals, and other external resources still require connectivity.

02 / DATA ARCHITECTURE

Two local services, two kinds of knowledge

The dashboard and Kiwix solve different problems. Separating them keeps the application small and lets each archive format use its native reader.



Request and search flow

User action	Path through the system
Open a category	Browser -> Python server /api/categories -> category folders under content/ -> response links to each file.
Search a phrase	Browser /api/search -> SQLite FTS5 query over path, title, category, and body -> up to 30 ranked snippets.
Open a PDF/HTML reference	Browser requests /content/... from the same local server; document is streamed from the SSD.
Read Wikipedia	Browser opens Kiwix on port 8081; Kiwix reads the .zim archive directly. Dashboard search does not index Wikipedia article bodies.
Check system status	Server reports mounted volume, total/free bytes, and local file counts/sizes.

Why keep the index derived?

The database is generated from content at startup. This avoids maintaining two sources of truth and makes it possible to restore search by copying app/ and content/ and letting the server rebuild index/library.sqlite3. The cost is a startup indexing pass, which can take several minutes on slower machines.

PDF limits: embedded text only, at most 100 MiB, 800 pages, and 1 million characters per PDF. Image-only scans need OCR. Map sheets are indexed by filename/source notes to avoid expensive parsing.

03 / BUILD SEQUENCE

From a dashboard sketch to a maintainable library

The implementation was incremental: first make the shell usable, then add the corpus and the capability needed to search it, then make startup portable.

Step	Work performed	Decision and rationale
1. Inspect and plan	Reviewed mounted volumes, available capacity, platform, and existing project state before writing to the SSD.	Keep existing data intact; use a dedicated project folder; do not format or repartition.
2. Build the dashboard	Created responsive HTML/CSS/JS cards, category navigation, local file links, and a status strip.	No frontend framework or build pipeline; the page can be served as local static assets.
3. Fix interaction issues	Adjusted the overlapping Explore link and added a way to close the side menu.	User-reported visual and navigation bugs were treated as concrete UX requirements.
4. Add Kiwix/Wikipedia	Added Kiwix Server and downloaded the English all-maxi ZIM, about 119 GiB.	Kiwix handles a large ZIM efficiently; this explains the overnight multi-hour download.
5. Add local search	Implemented SQLite FTS5 indexing and PDF text extraction with vendored pypdf.	Search stays local. ZIM browsing remains a separate Kiwix task.
6. Curate references	Added official/recognized references, local maps, Tacoma documents, and per-category source notes.	Prefer durable primary sources and record publisher, date, and URL; use static HTML when PDF downloads are blocked.
7. Add portable launchers	Packaged private Python and Kiwix runtimes for macOS ARM/Intel and Windows x64.	Per-platform launchers avoid requiring Python installation on each target computer.
8. Document handoff	Wrote project history, inventory, update procedures, and a future-agent preservation checklist.	A project should explain its why and operational limits, not just list files.

The multi-hour transfer

The long download was not a program installer. It was wikipedia_en_all_maxi_2026-08.zim, the offline English Wikipedia snapshot. Its SHA-256 sidecar is stored beside the archive.

04 / INFORMATION ARCHITECTURE

Curated by category, with provenance

The content tree is the library's data model: folder name determines category; filenames remain human-readable; SOURCES.md records where material came from.

Encyclopedia English Wikipedia maxi ZIM, 2026-08; ~119 GiB; checksum sidecar.	**Medical / first aid** CDC wound and preparedness documents; Red Cross first aid, burn, and bleeding guidance.
Wilderness NPS trip planning and outdoor emergency planning; USGS map/compass guide.	**Water** CDC water safety/storage references; EPA private-well guidance.
Food USDA home canning series; CDC outage safety; UGA storage; NCHFP drying/storage.	**Agriculture** NC central planting calendar; NC State gardening handbook, soil and seed saving.
Automotive 2017 Tacoma V6/4WD/automatic owner set; NHTSA tire reference; Toyota service link.	**Electrical** CPSC wiring hazards/safety; USFA electrical fire safety.
Plumbing / repair EPA septic guide; NC State preventative home maintenance.	**Generators / energy** CPSC and USFA generator safety; DOE renewable/solar resilience references.
Radio / communications FCC Part 97, ARRL ARES, IARU guide, NOAA weather radio and local NWS material.	**Maps / navigation** About 25 local, county, state and USGS PDFs around Albemarle, Harrisburg, Charlotte and nearby areas.
Preparedness / evacuation CDC preparedness documents, ReadyNC plan/evacuation/kit pages, household plan worksheet.	**Other references** Empty catch-all; populated later only when a resource does not fit a defined category.

Curation rule: for safety-critical references, prefer agencies, extension services, established safety organizations, and manufacturers. Where agencies block direct PDF retrieval, saved text pages can preserve the guidance offline; links to dynamic services remain online-only.

05 / PORTABILITY

One-click launch without installing Python

A conventional Python installer on the SSD would still require a separate installation on every computer. The selected approach is a private, compressed runtime unpacked to that computer's user cache.

Platform	Packaged runtime	Launch behavior
macOS Apple Silicon	CPython 3.14.10 arm64 archive; Kiwix Tools 3.8.2 arm64 executable already included.	Start Library.command selects arm64, extracts Python on first use to the user's Library cache, starts services, waits for HTTP readiness, opens browser.
macOS Intel	CPython 3.14.10 x86_64 archive; Kiwix Tools 3.8.2 x86_64 archive.	Same launcher selects Intel bundle; private runtime and Kiwix extract to an architecture-specific cache.
Windows x64	CPython 3.14.10 ZIP repacked from verified standalone upstream archive; Kiwix Tools 3.8.1 with ICU DLLs.	Start Library.bat calls PowerShell. The script expands runtimes under LOCALAPPDATA, starts Kiwix/server, waits for readiness, opens default browser.

Why archives instead of expanded folders on the SSD?

The SSD is exFAT. A trial copy of extracted portable runtimes consumed roughly 12 GiB because exFAT allocation overhead made thousands of small files expensive. The runtime trees were moved back into compact tar.gz/ZIP packages (~136 MB total) and are extracted locally on first run. This keeps the library portable without wasting SSD space.

Tradeoffs and boundaries

Benefit	Cost / limitation
No host Python or package manager needed; no system-wide install.	Each computer needs ~250 MB free local cache space; each architecture has its own package.
Mac Intel + Apple Silicon, Windows x64 packages stored offline.	Windows ARM is only conditional on x64 emulation. Windows build has not been exercised on a Windows device yet.
Component versions and archive checksums recorded.	Unsigned tools can trigger first-run security prompts; launchers are not a signed/notarized installer.

The bundled runtime is based on CPython from python-build-standalone; this is an application-private runtime, not a Python.org installer. The exact source release and hashes are in app/runtime/README.md.

06 / OPERATIONS

Large updates need staging and rollback

The same safe-update pattern applies to manuals and ZIM archives: stage outside the live content tree, verify, then switch while preserving the previous version.

Wikipedia is a complete snapshot

The present archive is wikipedia_en_all_maxi_2026-08.zim. Kiwix defines maxi as the full version; nopic trades away pictures to reduce size. The catalog says incremental ZIM updates are not currently available, so a new Wikipedia snapshot means a fresh whole-file download [1]. At the last check, about 762 GiB was free; check again before each download.

Step	Action
1. Select	Use the official Kiwix library/catalog. Confirm language, all-maxi variant, release date, exact size, and current checksum.
2. Stage	Download ZIM and checksum into backups/_incoming/, outside content/. Keep the current ZIM in place.
3. Verify	On Mac, use shasum -a 256 -c. On Windows, compare Get-FileHash -Algorithm SHA256 with publisher's value.
4. Stop	Stop the library launcher before moving a ZIM used by Kiwix.
5. Preserve	Move the old ZIM into backups/zim-archive/ or a second drive. Put the new dated archive and checksum in content/encyclopedia/.
6. Confirm	Restart; confirm category listing, open Kiwix, and load several articles. Keep rollback copy until verified.
7. Rebuild	The dashboard rebuilds its SQLite index at startup. It indexes the ZIM as a file entry, not Wikipedia article text.

Why preserve both copies?

A checksum detects transfer corruption, while keeping the old archive provides rollback if the new ZIM is incomplete or fails to open. The drive has room for a staged replacement at the last recorded capacity, but that must be checked again.

For smaller reference updates: prefer a primary source, record publisher/date/URL/license in the category SOURCES.md, use a dated filename, and restart to refresh search. Never stage partial downloads inside content/.

07 / ENGINEERING WORKFLOW

Agentic work is strongest when bounded by evidence

This project combined human direction with incremental implementation. The workflow is useful beyond this particular library.

Practice	How it was applied
Inspect before changing	Confirmed mounted project path, existing app/data, filesystem notes, free capacity, and current service endpoints before writing to the SSD.
Separate code from data	Kept app/, content/, index/, backups/, runtime packages, and source notes distinct.
Let feedback steer implementation	Fixed concrete UX issues and then added features requested by the owner, such as searchable PDF text and a Tacoma-specific collection.
Use verifiable sources	Downloaded official references, recorded URLs/check dates in SOURCES.md, verified runtime/archive checksums, and preserved older copies.
Prefer staged, reversible changes	New files go to a staging folder; replacement is versioned; old data is retained; no formatting/partitioning/erasing.
Report limits honestly	Distinguish local text from live links; note OCR, map-rendering, missing factory repair manual, and the untested Windows launch.

Tools and responsibilities

Tool	Job in this build
Python 3 / http.server	Serve local dashboard and content; build the search index with built-in SQLite.
SQLite FTS5 + pypdf	Index text fields and extract embedded PDF text without cloud search or a separate database server.
Kiwix Tools	Serve ZIM archives using their native reader and local HTTP endpoint.
Bash, PowerShell, batch	Platform launchers select runtimes, start processes, wait for readiness, open the browser, and manage shutdown.
curl, tar, unzip, shasum, MD5	Retrieve packages, unpack for inspection, and verify integrity before bundling.
Safari/Chrome	Local UI review via loopback URLs; no frontend bundler or node dependency at runtime.

Project process lesson: a good agentic handoff states the invariant (preserve user data), shows current evidence (paths, sizes, versions), separates reversible work from approvals, and leaves source and rollback instructions alongside the artifact.

08 / STATUS AND NEXT STEPS

What is ready, what still needs a real-device check

This is a working offline library, not a finished emergency platform. The following distinction prevents a polished dashboard from implying broader coverage than the corpus actually has. Both launchers now explicitly restrict Kiwix to loopback; an older running process must be restarted to apply that setting.

Area	Current state
Mac runtime	Apple Silicon Python 3.14.10 runtime smoke-checked for SQLite FTS5 and pypdf import. Current dashboard API returned status successfully during documentation work.
Package integrity	Python archives checked against SHA-256 values; Kiwix Intel Mac and Windows packages checked against publisher MD5 values. The Windows Python ZIP has its own recorded SHA-256.
Windows device launch	Not yet executed on a physical Windows PC. First real Windows use should be treated as a deployment smoke check; report any PowerShell, SmartScreen, or Kiwix issue.
Intel Mac launch	Architecture-specific runtime and Kiwix package are present, but the Intel package was not exercised on Intel hardware.
Network exposure	Python dashboard binds to 127.0.0.1. Kiwix defaults to all interfaces; both launchers now pass --address=127.0.0.1. A process started before that change must be restarted.

Known product gaps

No interactive offline map renderer; no OCR for scanned PDFs; no full Toyota factory repair manual; no live alerts or shelter data; some categories remain thinner than their descriptions. The catch-all Other category is empty. Medical and emergency material is reference content and can become stale.

Recommended next work

1. Run the Windows launcher on a real x64 Windows 10/11 computer and verify dashboard, PDF search, and Kiwix. 2. Decide whether Kiwix should be explicitly restricted to loopback. 3. Expand home repair and electrical safety content from appropriate primary sources. 4. If maps must be interactive, select a local map viewer and offline data format; do not imply the current PDF collection is live navigation.

Selected references

[1] Kiwix catalog FAQ - https://get.kiwix.org/en/solutions/catalog/
Variants and lack of incremental ZIM updates.

[2] Kiwix Wikipedia ZIM index - https://download.kiwix.org/zim/wikipedia/
Current names, dates, and archive sizes.

[3] Kiwix Tools source repository - https://github.com/kiwix/kiwix-tools
Server options, release source, and GPL license.

[4] python-build-standalone release - https://github.com/astral-sh/python-build-standalone/releases/tag/20261026
Private CPython runtime archives bundled here.

[5] Microsoft Windows 10 support status -
https://support.microsoft.com/en-us/windows/deployment/updates-lifecycle/windows-10-support-has-ended-on-october-14-2025
App compatibility is separate from operating-system security support.

[6] NC State Extension vegetable gardening - https://content.ces.ncsu.edu/extension-gardener-handbook/16-vegetable-gardening
Example regional primary source for gardening references.

[7] CDC emergency water supply - https://www.cdc.gov/water-emergency/about/how-to-create-and-store-an-emergency-water-supply.html
Example public-health source and water-storage reference.

For detailed source citations for the other references, see content/<category>/SOURCES.md on the SSD. For the exact launch commands, file inventory, and safe update procedure, see PROJECT HANDOFF.md and UPDATE GUIDE.md in the library root.